

SPIKE ESSENTIAL



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
16 CLASSES
(1 HOUR PER CLASS)**

**MODE:
OFFLINE**

LEGO® SPIKE™ Essential is an integrated hardware and software platform designed to introduce students to robotics, engineering, and coding through hands-on model construction and programming.

Students build models step-by-step using official LEGO instructions, understand how each mechanism functions, and then program their creations using a visual block-based interface.

Through this process, students learn:

- Mechanical design principles
- Real-world industrial components (pulleys, cams, axles, gears, etc.)
- Logical and sequential programming structures
- Basic mathematical operations within coding

The learning experience becomes more engaging when students program their models using drag-and-drop coding blocks to control motors, sensors, lights, and sounds.

COURSE CURRICULUM

Kit Components & Features

Hardware Components

- **2 Ports**
Used to connect motors and sensors to the Hub.
- **Color Sensor**
Detects specific colors and measures light intensity to trigger programmed actions.
- **Light Matrix (3×3 LED Display)**
Displays patterns, numbers, symbols, or animations to provide visual feedback.

Mechanical Parts

- **Motors**
Provide motion and power to wheels or other mechanical systems.
Can rotate at specific speeds, angles, or durations.
- **Wheels and Axles**
Wheels enable smooth movement.
Axles transmit rotational motion between motors and wheels.
- **Gears**
Transfer rotational motion.
Adjust speed and torque using gear ratios.
Enable precision control in robotic mechanisms.
- **Pulleys**
Transfer force using belts.
Create lifting systems such as cranes or elevators.
- **Beams and Connectors**
Provide structural strength.
Form the framework of robotic models.

Programming Platform

SPIKE Essential uses an intuitive block-based coding environment that allows students to program logically and sequentially.

1. Motor Blocks

- **Start Motor** – Activates motor at a defined speed and direction.
 - **Stop Motor** – Stops motor immediately.
 - **Set Speed** – Adjusts motor speed.
 - **Move for Rotations/Seconds** – Runs motor for a specified time or rotation count.
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2. Light Blocks

- **Set Light Color** – Changes the color of the hub's light matrix.
 - **Blink Light** – Flashes LED display.
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3. Sound Blocks

- **Play Sound** – Plays built-in sound effects.
 - **Play Note** – Produces musical notes for a defined duration.
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4. Event Blocks

- **When Program Starts** – Initiates program execution.
 - **When Button Pressed** – Triggers actions on button input.
 - **When Tilted** – Detects hub orientation and executes actions.
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5. Control Blocks

- **Wait For** – Pauses execution until condition is met.
 - **Repeat Loop** – Repeats instructions a set number of times.
 - **Forever Loop** – Runs code continuously.
 - **If-Else** – Executes conditional logic.
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6. Sensor Blocks

- **Color Sensor Block** – Reads color and light intensity values.
- **Hub Tilt Sensor Block** – Detects orientation and motion.